

Priyanshu Patra

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EDUCATION

- **ARIZONA STATE UNIVERSITY, TEMPE, AZ, USA** *Aug 2025 – Present*
Master of Science in Data Science, Analytics and Engineering (DSAE) | GPA: **4.0/4.0**
 - Relevant Coursework: Statistics for Data Analysts, Data Processing at Scale, Information Assurance and Security, Data Visualization, Statistical Machine Learning, Data Mining.
- **SRM INSTITUTE OF SCIENCE AND TECHNOLOGY, CHENNAI, INDIA** *Sep 2020 – May 2024*
Bachelor of Technology (B. Tech) Major in Computer Science and Engineering (CSE) | GPA: **3.58/4.0**
 - Relevant Coursework: Data Structures and Algorithms, Database Management Systems (DBMS), Object-Oriented Programming (OOP), Computer Networks, Artificial Intelligence & Machine Learning, Web Development.

PROJECTS & INDUSTRY COLLABORATIONS

- MA Rank - Multi Agent Resume Screening System | Python, LangGraph, Neo4j, NLP, Streamlit** *Mar 2026 – May 2026*
 - Built a multi agent LLM pipeline using LangGraph with five specialized agents for resume parsing, skill normalization, graph storage, ranking and consensus evaluation stored in Neo4j knowledge graph.
 - Developed composite scoring algorithm combining skill coverage, semantic similarity and experience fit achieving Precision at 10 of 0.80 outperforming BM25 baseline of 0.50 across 500 resumes and 1534 job postings.
 - Deployed recruiter facing Streamlit dashboard enabling real time resume ingestion, candidate ranking and explainable shortlisting with matched skills, missing skills and uncertainty scores.
- Urban vs Rural Energy Intensity Analysis | Python, Pandas, Statsmodels, GeoPandas, Matplotlib** *Oct 2025 – Dec 2025*
 - Merged NREL and USDA government datasets using county FIPS codes engineering a composite Energy Intensity Index across all 15 Arizona counties revealing rural counties exhibit nearly double urban per capita energy burden.
 - Built two OLS regression models identifying transportation fuel as dominant energy intensity driver achieving R squared of 1.000 versus urban classification baseline of 0.052.
 - Generated choropleth maps and statistical visualizations translating findings into actionable policy recommendations targeting rural EV infrastructure and transportation efficiency improvements.
- AI-Powered Resume Screening System | Edunet x Microsoft & SAP** *Feb 2025 – Mar 2025*
 - Built an NLP-based resume screening pipeline using Python and spaCy (NLP library), improving candidate ranking accuracy and reducing manual review time by ~40%.
 - Deployed screening model as a Flask REST API, enabling real-time resume scoring with sub-300ms response time across concurrent requests.
 - Achieved 87% classification accuracy on a labeled dataset of 1,200+ resumes, validated through cross-validation and precision-recall analysis.
- Finance Tracker Full-Stack App | Edunet x EY-GDS** *Sep 2024 – Nov 2024*
 - Built a real-time expense tracking application using the MERN stack, implementing Recharts for data visualization and reducing manual entry errors for beta users.
 - Executed JSON Web Token (JWT)-based authentication and role-based access control, ensuring secure multi-user data isolation across AWS-hosted deployment.
- OCR Food Allergen Detection | Python, Tesseract, OpenCV, Generative AI** *Dec 2023 – May 2024*
 - Automated allergen data extraction from product labels using Tesseract and OpenCV, achieving 94% accuracy and enabling real-time compliance flagging for 200+ images/min.
 - Deployed detection pipeline on Google Cloud Platform (GCP), processing 200+ product images per minute with automated allergen flagging for compliance use cases.

SKILLS

- Programming:** Python, Java, JavaScript, C, SQL
- Data Science & ML:** Pandas, NumPy, Scikit-learn, TensorFlow, NLP, Statistical Modeling, Recommendation Systems, Churn Analysis, A/B Testing, Power BI, Google Cloud Platform (GCP), PyTorch, PySpark, Hadoop, Excel, Tableau
- Web & Backend:** Flask, Django, React.js, Node.js, HTML, CSS
- Databases & DevOps:** MySQL, MongoDB, Docker, Advanced SQL, Kubernetes, Database Automation Scripting
- Cloud & Tools:** AWS, Microsoft Azure, GitHub, MATLAB
- Mathematics & Optimization:** Linear Algebra, Calculus, Optimization Theory, Time Series Analysis, Hypothesis Testing

CERTIFICATIONS

- AWS Educate Introduction to Generative AI by Amazon Web Services** *April 2026*
- Microsoft AI & ML Engineering by Coursera** *Mar 2025*
- SQL (Advanced) by HackerRank** *Mar 2025*
- Problem Solving (Intermediate) by HackerRank** *Mar 2025*
- Oracle Database Foundations by Oracle** *Mar 2023*